Adopted Levels

Type	Author	History Citation	Literature Cutoff Date
Full Evaluation	Balraj Singh	NDS 156, 70 (2019)	31-Jan-2019

S(n)=7480 *SY*; S(p)=2210 *SY*; Q(α)=11861 *30* 2017Wa10,2018Br13
S(n) and S(p) from 2017Wa10. Q(α) from Eα value in 2018Br13.
Estimated uncertainties (2017Wa10): ΔS(n)=970, ΔS(p)=1050.
S(2p)=3360 *1040*, Q(εp)=1510 *890* (syst,2017Wa10). S(2n)=14070 (1997Mo25,theory).
Q(α) is from Eα=11.70 MeV *3* (2018Br13, from analysis of a total of five decay chains observed for the decay of ^{294}Og in different experiments. Other: 11840 keV *70* (2017Wa10).
^{294}Og observed only in experiments at FLNR-JINR-Dubna, in collaboration with LLNL and ORNL laboratories in USA.
2006Og05, 2004Og10, 2004Og12 (also 2009Og07,2007Og05,2005Og03, 2005OgZZ,2004Og05,2003OgZZ,2002OgZX): discovery of Z=118 (Og) element through the identification of ^{294}Og nuclide decaying by three successive α-decay chains: ^{290}Lv, ^{286}Fl and ^{282}Cn, which decays by SF. ^{294}Og from ^{249}Cf(^{48}Ca,3n),E=251 MeV; >98% enriched ^{249}Cf rotating target. The compound nucleus ^{297}Og formed at 32.1-36.6 MeV. Facility: U400 cyclotron with the Dubna gas-filled recoil separator (DGFRS) of the Flerov Laboratory of Nuclear Reactions (FLNR), Joint Institute for Nuclear Research in Dubna (JINR), in collaboration with LLNL, USA. Evaporation residues recoiling from the target were separated by the DGFRS in flight from ^{48}Ca beam ions, transfer reaction products and scattered particles. Multiwire proportional counter for TOF measurement; semiconductor focal-plane detector array with 12 vertical position-sensitive strips surrounded by eight side detectors without position sensitivity. The α particles were measured in the range of 9.9 MeV to 12.0 MeV. FWHM=70-120 keV for α particles absorbed in the focal-plane detector, 280-410 keV for α particles that escaped the focal-plane detector and were registered by the side detector.
History of three events observed: two correlated decay chains detected in the experiment at E(^{48}Ca=251 MeV) reported by 2006Og05, and one correlated decay chain in an earlier experiment (2004Og10) at E(^{48}Ca=245 MeV). 2004Mo15 pointed out that the mass assignment in their work may be off by one unit. 2005St38 analyzed probability of observing a random event in their work.

Event #1: Energy of the evaporation residue=13.2 MeV (2004Og10).
$E_{\alpha1}$=11650 *60*, $\Delta t_{\alpha1}$=2.549 ms, assigned to ^{294}Og.
$E_{\alpha2}$=10710 *170*, $\Delta t_{\alpha2}$=42.1 ms, assigned to ^{290}Lv.
E_{SF}=207 MeV, Δt_{SF}=517.6 ms, assigned to ^{286}Fl SF decay.

Event #2: Energy of the evaporation residue=13.7 MeV (2006Og05).
$E_{\alpha1}$=11800 *500*, $\Delta t_{\alpha1}$=0.847 ms, assigned to ^{294}Og.
$E_{\alpha2}$=10800 *90*, $\Delta t_{\alpha2}$=0.098 ms, assigned to ^{290}Lv.
$E_{\alpha3}$=10160 *90*, $\Delta t_{\alpha3}$=153.0 ms, assigned to ^{286}Fl; decays by α and 70% by SF mode.
E_{SF}=202 MeV, Δt_{SF}=2.70 ms, assigned to ^{282}Cn SF decay.

Event #3: Energy of the evaporation residue=10.4 MeV (2006Og05).
$E_{\alpha1}$=11650 *100*, $\Delta t_{\alpha1}$=0.465 ms, assigned to ^{294}Og.
$E_{\alpha2}$=10840 *100*, $\Delta t_{\alpha2}$=1.012 ms, assigned to ^{290}Lv.
E_{SF}=157 MeV, Δt_{SF}=11.39 ms, assigned to ^{286}Fl SF decay.

2012Og06 (also 2011Og07): ^{249}Cf(^{48}Ca,3n),E=251.7 MeV; ^{249}Cf from decay of ^{249}Bk target; rotating ^{249}Bk radioactive target. ^{249}Bk radioactive target was produced in neutron irradiation of Cm and Am targets in high flux isotope reactor at ORNL. The chemically purified 12.7 mg of ^{249}Bk (about half of the total ^{249}Bk material produced at ORNL) containing 0.51 mg of ^{249}Cf with no other detectable impurities was shipped to JINR. Six targets were prepared at RIAR facility in Russia, each target with a thickness of 0.33 mg/cm^2 deposited on 0.72 mg/cm^2 thick Ti foils, and mounted on a disk that was rotated at 1700 rpm perpendicular to the beam direction. U-400 cyclotron with the Dubna gas-filled recoil separator (DGFRS) of the Flerov Laboratory of Nuclear Reactions, Joint Institute for Nuclear Research in Dubna (JINR). Evaporation residues passing through the DGFRS separator registered by time-of-flight system. Focal plane detector system: two 6x6 cm^2 Si detectors, each of 16 strips with position sensitivity in the vertical direction. These detectors were surrounded by six similar 6x6 cm^2 Si side detectors without position sensitivity. Behind the focal plane detectors, two veto Si detectors, similar to the side detectors, were mounted for detecting or rejecting signals from high-energy long-range charged particles produced in direct reactions of ^{48}Ca with the DGFRS materials.

Adopted Levels (continued)

Measured Eα, half-lives, α-α correlations. FWHM for α particles was 34-73 keV. Collaboration: JINR and RIAR, Russia; LLNL, ORNL, University of Tennessee and Vanderbilt university.

One event was assigned to the decay of ^{294}Og in 2012Og06; two events were assigned earlier in 2006Og05 and one in 2004Og10.

Event #4 (2012Og06):
Excitation energy of the compound nucleus ^{297}Og=11.63 MeV.
$E_{\alpha1}$=11.679 MeV *65*, $\Delta t_{\alpha1}$=0.135 ms, assigned to ^{294}Og.
$E_{\alpha2}$=10.848 MeV *65*, $\Delta t_{\alpha2}$=29.437 ms, assigned to ^{290}Lv.
E_{SF}=219.3 MeV, Δt_{SF}=3.486 ms, assigned to ^{286}Fl, decays by α and SF.

2018Br13: E(^{48}Ca)=252,258 MeV beam from U400 cyclotron of JINR-Dubna was incident on a mixed Cf target (50.6% ^{249}Cf, 13.1% ^{250}Cf and 36.3% ^{252}Cf) obtained from >30 year old ^{252}Cf samples at ORNL. The Cf target was fabricated at ORNL as 12 segments, each with a layer of 309 μg/cm^2 thick Cf, containing a total of 1 mg of mixed Cf source. Fragments were separated by Dubna Gas-filled recoil separator (DGFRS), and ions were tagged with time-of-flight (TOF) in two pentane filled multi-wire proportional counters (MWPC). Focal plane detector array for the detection of α particles, designed and assembled at ORNL, was a double-sided silicon detector (DSSD) consisting of 48 horizontal strips on front side and 128 vertical strips on the back side for position resolution. The escaped α particles were detected in a box of six Si detectors surrounding the focal plane detector. A veto detector was placed behind the focal plane detector to reject light particles. Calibration of the detector system and assessment of target condition was carried out using Yb(^{48}Ca,xn) reaction producing α-decaying isotopes of ^{216}Th to ^{220}Th. Measured recoil-α correlated events, and production cross section. Deduced one event assigned to the decay of ^{294}Og through ^{294}Og → ^{290}Lv → ^{286}Fl. Reevaluated $T_{1/2}$ of ^{294}Og decay, and Eα emitted in this decay by combining the previous data for four observed events and the one in the present work. Comparison with several theoretical predictions of half-lives of α decay of ^{294}Og and its Q(α).

Event #1: Energy of the evaporation residue=6.315 MeV (2018Br13).
$E_{\alpha1}$=11717 *30*, $\Delta t_{\alpha1}$=0.185 ms, assigned to ^{294}Og.
$E_{\alpha2}$=10861 *140*, $\Delta t_{\alpha2}$=9.026 ms, assigned to ^{290}Lv.
E_{SF}=181.8 MeV, Δt_{SF}=119.497 ms, assigned to ^{286}Fl SF decay.

For theoretical studies, consult Nuclear Science References (NSR) database at NNDC, BNL for 143 primary references dealing with the half-lives, fission barriers and other aspects of nuclear structure in this mass region.

^{294}Og Levels

E(level)	J^π	$T_{1/2}$	Comments
0	0^+	0.58 ms *+44−18*	%α=100 (2016Ho09); %SF≤20 (2016Ho09) Only the α decay mode has been observed. %SF≤20 (2016Ho09), ≤50 (2006Og05). 2018Br13 searched for SF decay mode of ^{294}Og, but presence of a large number of SF events, possibly from SF decay of ^{258}No, obscured any SF activity from ^{294}Og. E(level): observed activity is assumed to correspond to the g.s. of ^{294}Og. $T_{1/2}$: from evaluation by 2018Br13 based on observation of a total of five decay chains for ^{294}Og in different experiments). one observed by 2018Br13, one by 2012Og06, two by 2006Og05, and one by 2004Og12. Others: 2012Og06 (also 2017Og01 and 2015Og05 reviews) gave $T_{1/2}$=0.69 ms *+64−22* ms from four observed events. Value of 0.69 ms *+69−23* is listed in 2016Ho09 review. Earlier values: 0.89 ms *+107−31* (from three events,2006Og05,2011Og07); 1.8 ms *+750−13* (2004Og12, from one event at 95% confidence level, or 1.8 ms *+84−8* at 68% confidence level). Cross section for production=0.9 pb *+32−8* (2018Br13) at E^*=35.2 MeV *22* and 36.4 MeV *22*; 0.30 pb *+70−26* (2012Og06) at E^*=26.6-37.5 MeV; and 0.5 pb *+16−3* (2006Og05) at E^*=32.1-36.6 MeV, where E^* is the excitation energy of the compound nucleus of ^{297}Og. Eα=11.70 MeV *3* (evaluated by 2018Br13 from five observed decay chains, one in their experiment, and four in previous experiments, as mentioned above in comment for $T_{1/2}$). Others: 11.66 MeV *6* (2012Og06, from four events); 11650 *60* (2006Og05, 2011Og07 from three events).